30. THE MALE GENITAL SYSTEM

DURATION : 06:40 STUDY SHEET

COURSE SUMMARY

This video deeply explores the formation of the male reproductive system through different embryonic stages. You will learn how the genital ridge, influenced by the migration of gonocytes, gives rise to the gonads, as well as the critical role of the Wolffian ducts, which transform into structures such as the vas deferens and epididymis, while the Müllerian duct regresses. The video also details the mechanism of testicular descent, highlighting the importance of the gubernaculum ligament and its interaction with liver development. Finally, it addresses the prostate, its embryological origins, and its connection to the urogenital sinus, providing a holistic understanding necessary for practitioners and students in embryology and osteopathy.

THE MALE GENITAL SYSTEM

1. THE GENITAL RIDGE AND GONAD FORMATION

The **genital ridge** is a structure influenced by the migration of **gonocytes**, which originate from the **yolk sac**.

These gonocytes thicken the posterior coelomic epithelium, thus forming the anlage of the gonad, the future **ovary** or **testis**. This tissue is swollen by the proliferation of the gonocytes that reach it.

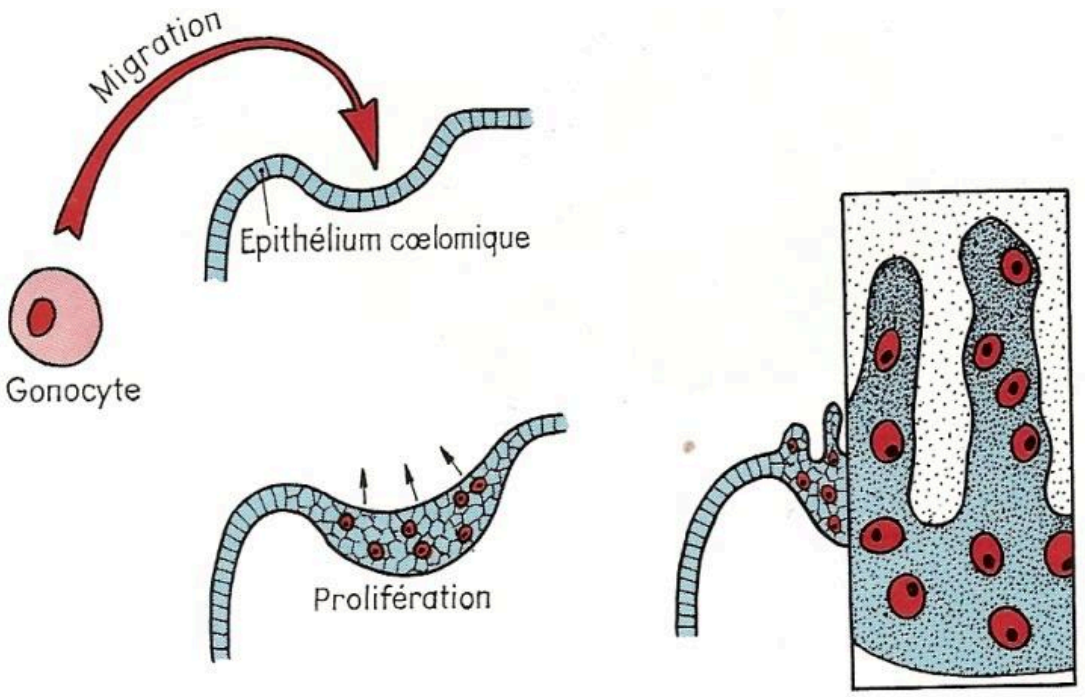


Fig. 6. — *Diagramme illustrant la migration gonocytaire.*

FIG. — Gonocyte migration

2. THE ROLE OF THE WOLFFIAN AND MÜLLERIAN DUCTS

The **Wolffian duct** is a collecting duct, already formed and then disappeared since the **pronephros**. It is pushed laterally and a small duct, the **Müllerian duct**, appears.

These two ducts, Müllerian and Wolffian, play a crucial role in the development of the **urogenital** system, in both males and females.

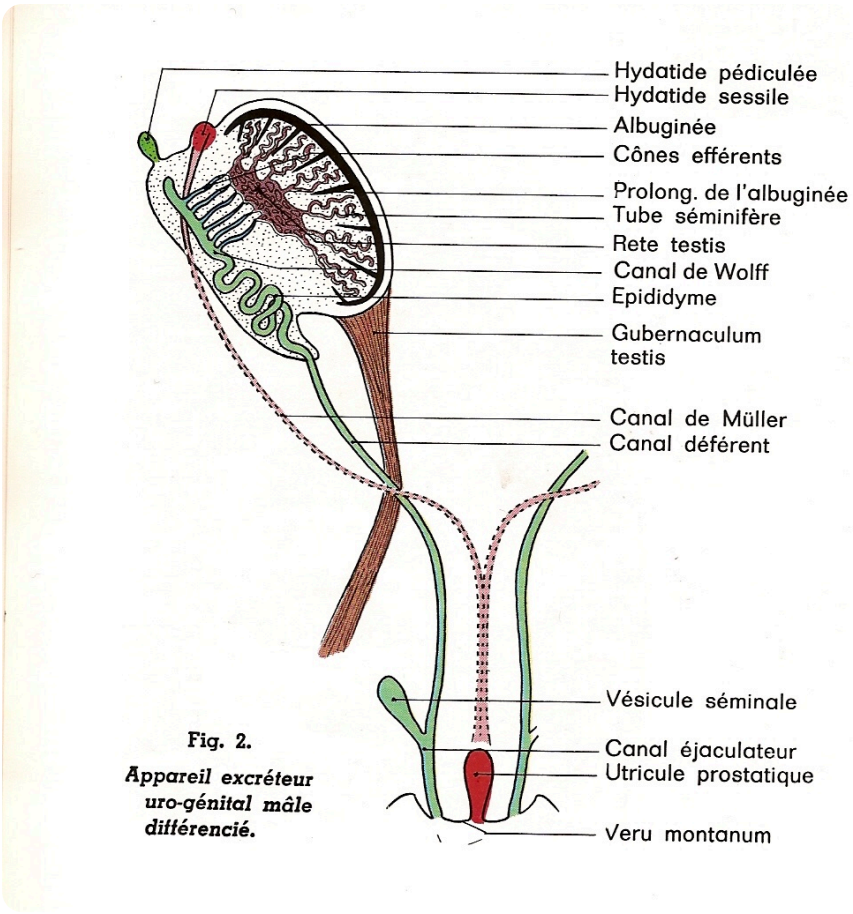


FIG. — General role of Wolffian and Müllerian ducts, in connection with the section introducing them

2.1. DEVELOPMENT IN MALES

In males, the Wolffian duct differentiates into the **vas deferens** and, by budding, into the **ureter**. It connects to the developing testicular area, forming connections that will give rise to the **epididymis**.

The Müllerian duct, on the other hand, regresses in males, leaving behind a small vestige, the **sessile hydatid**, which has no known function. However, it gives rise to the **prostatic utricle**, which will later be engulfed by endodermal tissue to form the **prostate**. This prostate will be connected to the system of vasa deferentia, originating from the Wolffian duct.

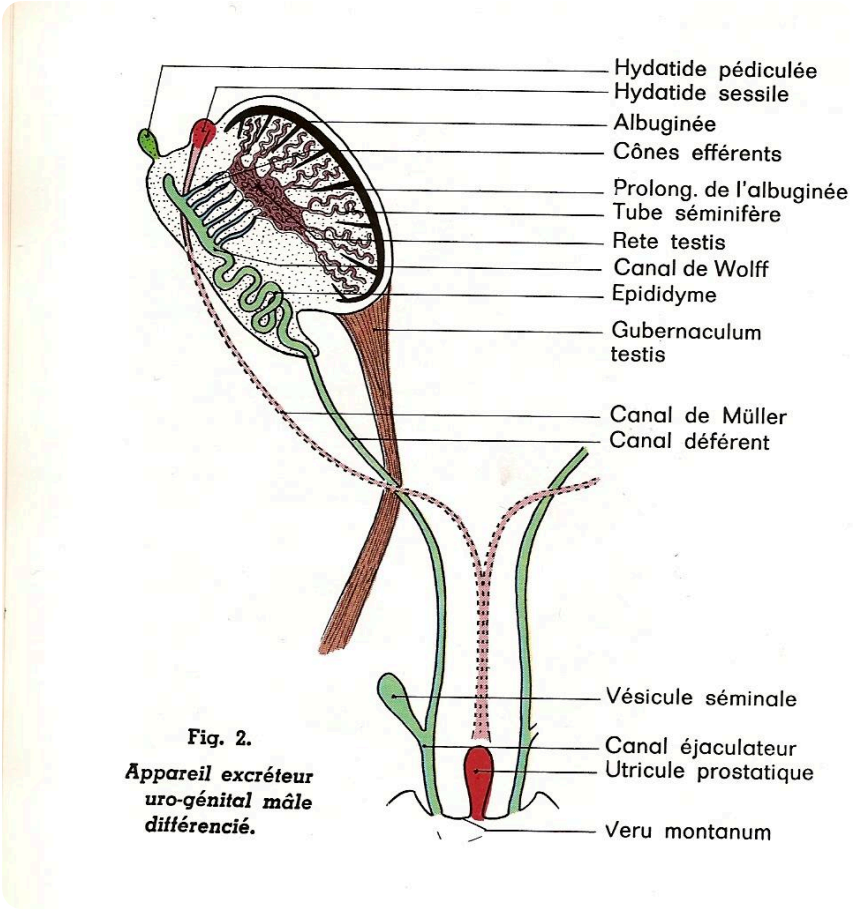


FIG. — Differentiated male genitalia

2.2. DEVELOPMENT IN FEMALES

In females, the Müllerian duct forms the **fallopian tubes** and the **uterus**.

3. THE GUBERNACULUM LIGAMENT AND TESTICULAR DESCENT

In males, a crucial ligament is the **gubernaculum testis**. It shortens and pulls the testes downwards.

This process is linked to the elongation of the rest of the system and the differential growth rates of the posterior part. Thus, the gonad, initially in a high position, descends. The gubernaculum, associated with the **inguinal ligament**, draws the testicular system downwards.

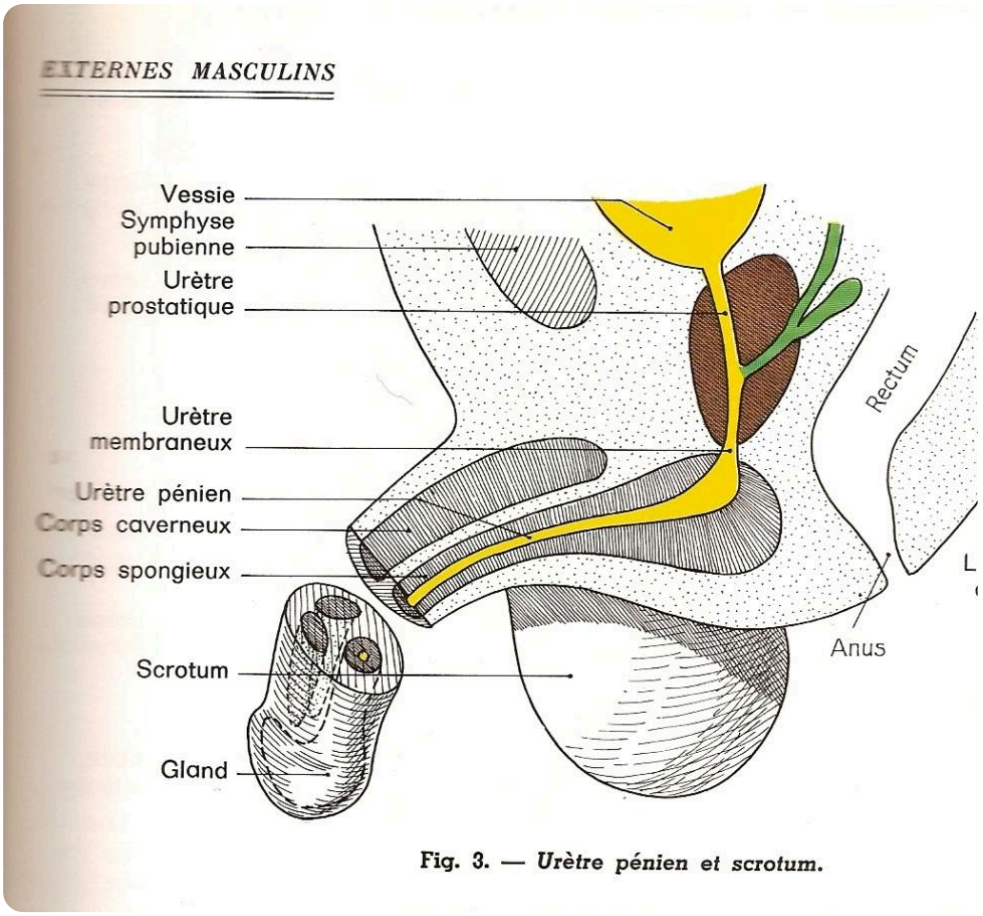


FIG. — Male reproductive system anatomy

4. THE INFLUENCE OF HEPATIC MOVEMENT

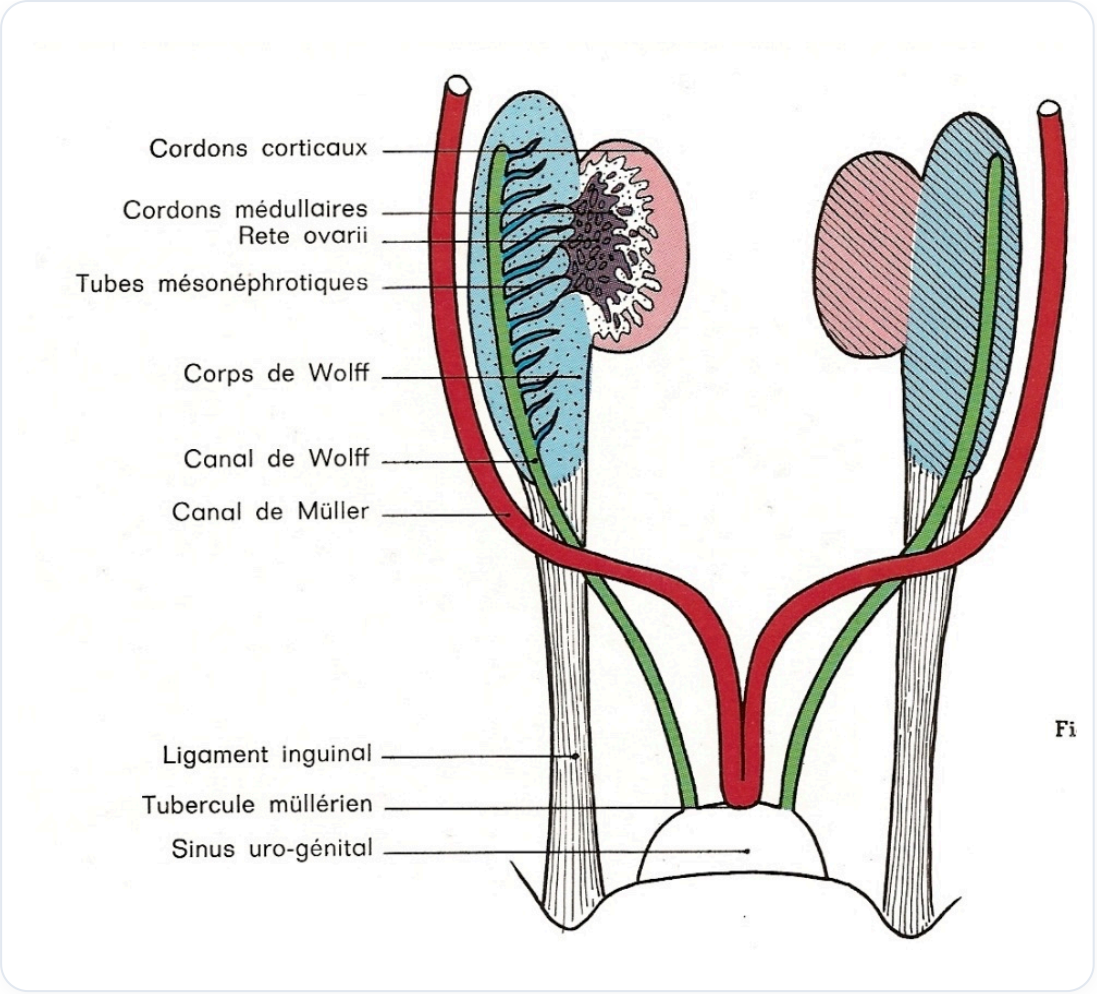


FIG. — Testicular descent, a key concept explained in this paragraph

The initial **hepatic movement** pushes the whole system back and organises the peritoneal cavity as well as urogenital development.

The liver therefore has major importance, both in its metabolic and embryonic functions.

5. THE PROSTATE AND THE UROGENITAL SINUS

The **prostatic utricle** is a remnant of the Müllerian duct, composed of urogenital epithelial tissue of endodermal origin.

The prostate develops from this tissue. The core of the prostate (the prostatic utricle) is **mesenchymal**, while its capsule is endodermal, originating from the **urogenital sinus**.

The urogenital sinus is the area where the **cloaca**, the **allantois** and, further back, the **rectum** were located. It is also from here that the **ureteric bud** originates, which induces the definitive kidney, itself derived from the Wolffian duct.

6. DETAILED TESTICULAR DESCENT

The initial position of the testes is very high. Testicular descent is due to the non-growth of the gubernaculum ligament, which pulls the system by the surrounding growth.

Meanwhile, the other ligament, the **gonadal suspensory ligament** (formerly called the ovarian suspensory ligament, but more precisely the **testicular suspensory** or **diaphragmatic** ligament), lengthens.

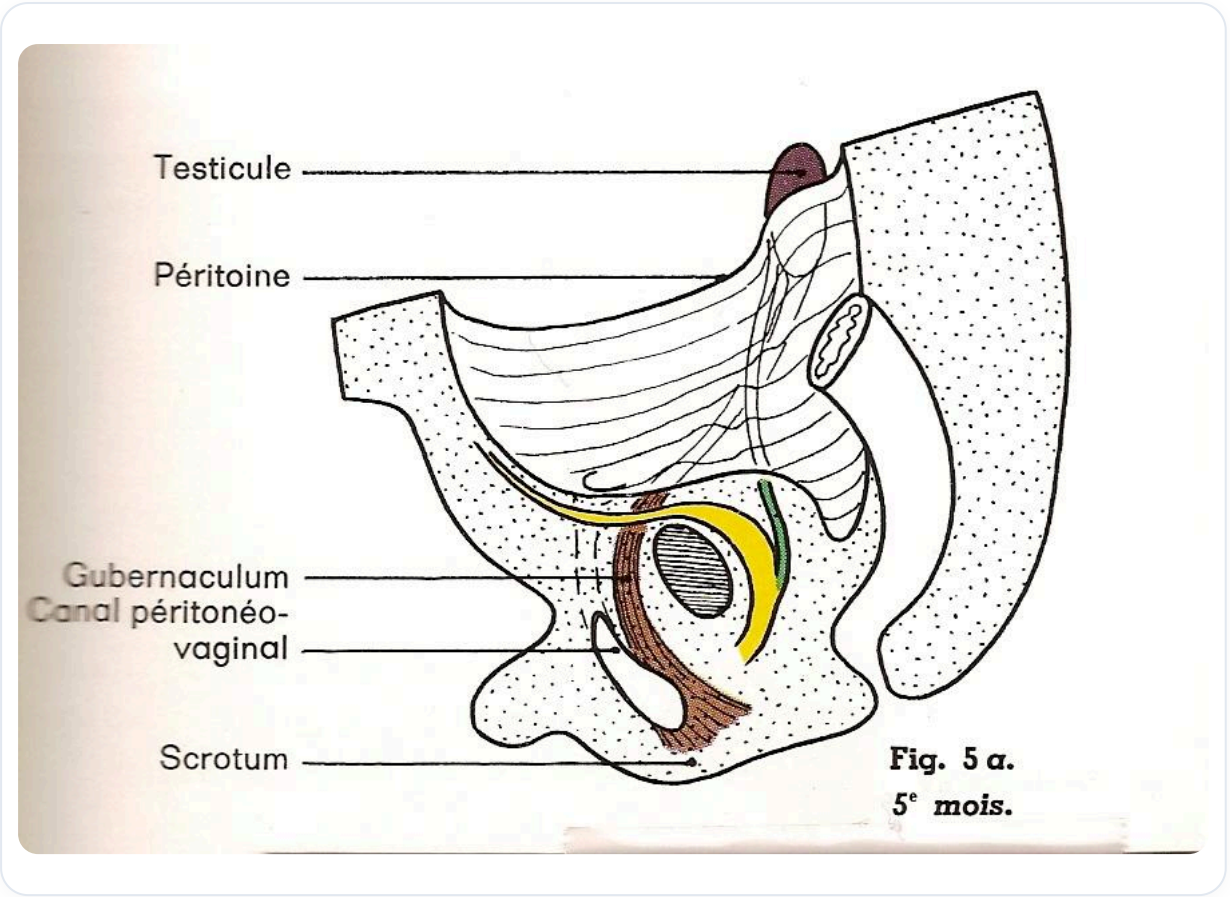


FIG. — Fetal testicular descent

This attachment between the gubernaculum ligament and the lower layers, combined with differential growth, causes testicular descent. It is in fact an overall growth that gives the impression of a reduction in the ligament, whereas it is the rest that grows.

The initial suspension of the testes up to the diaphragmatic plane, and more precisely to the **duodeno-pancreatic** junction, is a crucial point to review.

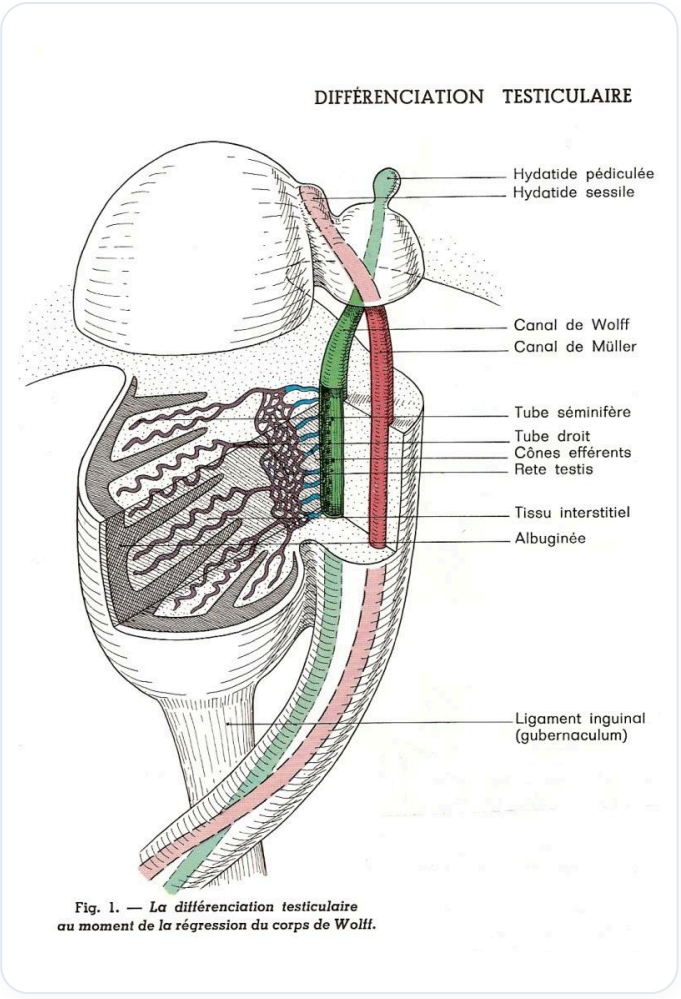
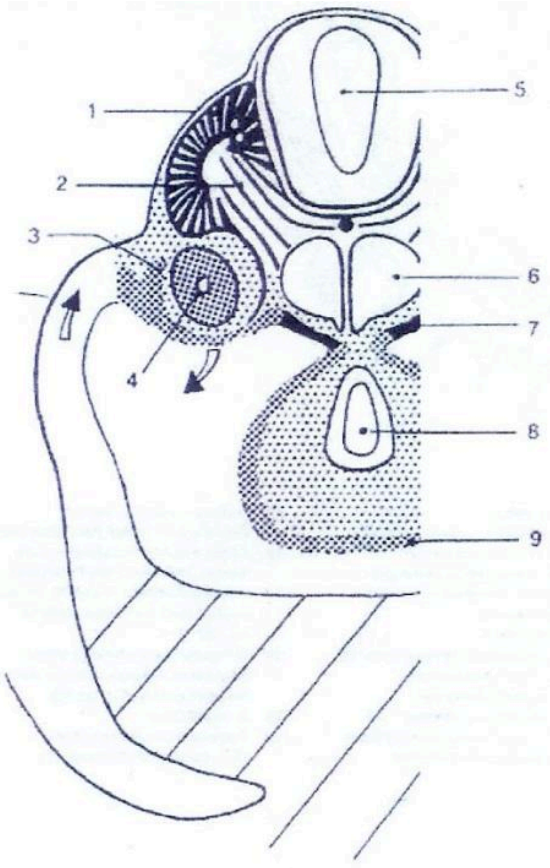


Fig. 1. — *La différenciation testiculaire au moment de la régression du corps de Wolff.*

FIG. — Testicular differentiation



Embryo, age approximately 27 days and 2,7 mm

1. Dermatom
2. Skelrotom
3. Ductus of wolff
4. Inner part of mesonephros
5. Neural tube
6. Aorta
7. Gonad
8. Intestinal tube
9. Limiting tissue body cavity

FIG. — Embryo 27 days

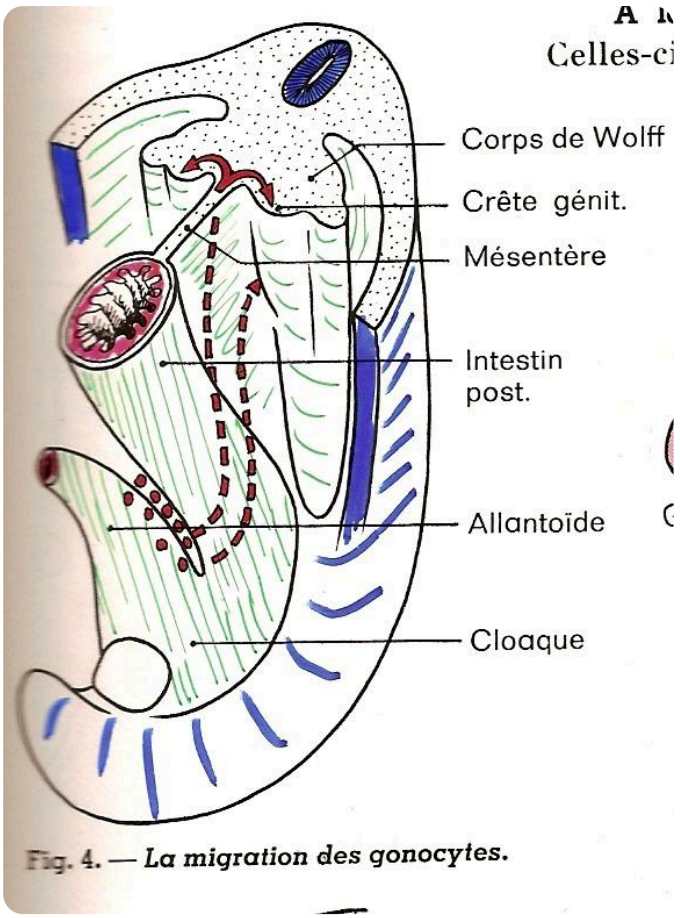


Fig. 4. — *La migration des gonocytes.*

FIG. — Migration of gonocytes

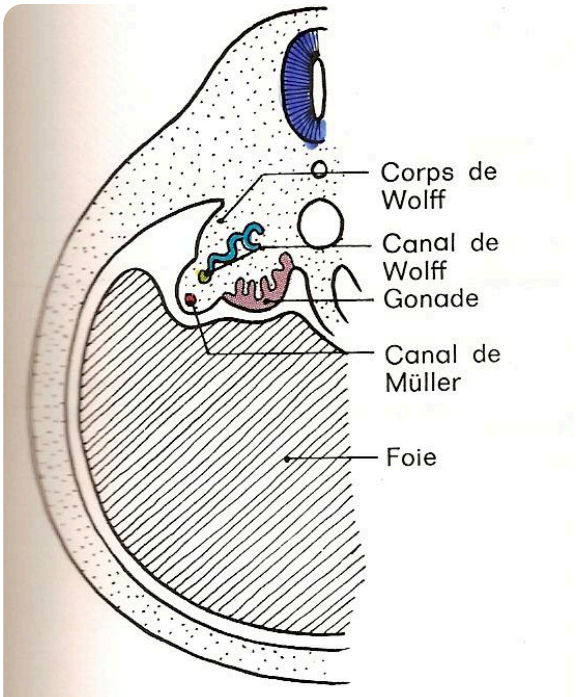


Fig. 5. — *La gonade indifférenciée au moment de l'apparition des cordons sexuels.*

FIG. — Undifferentiated gonad

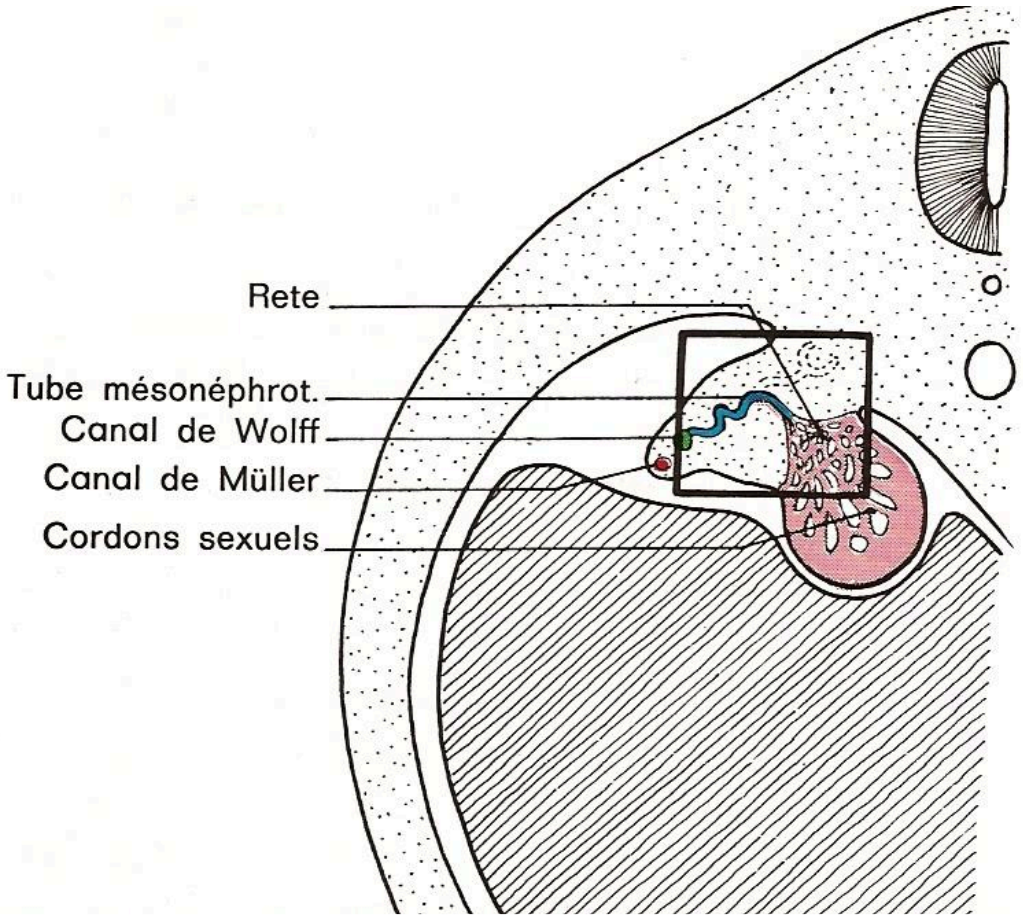


FIG. — Embryonic gonadal differentiation

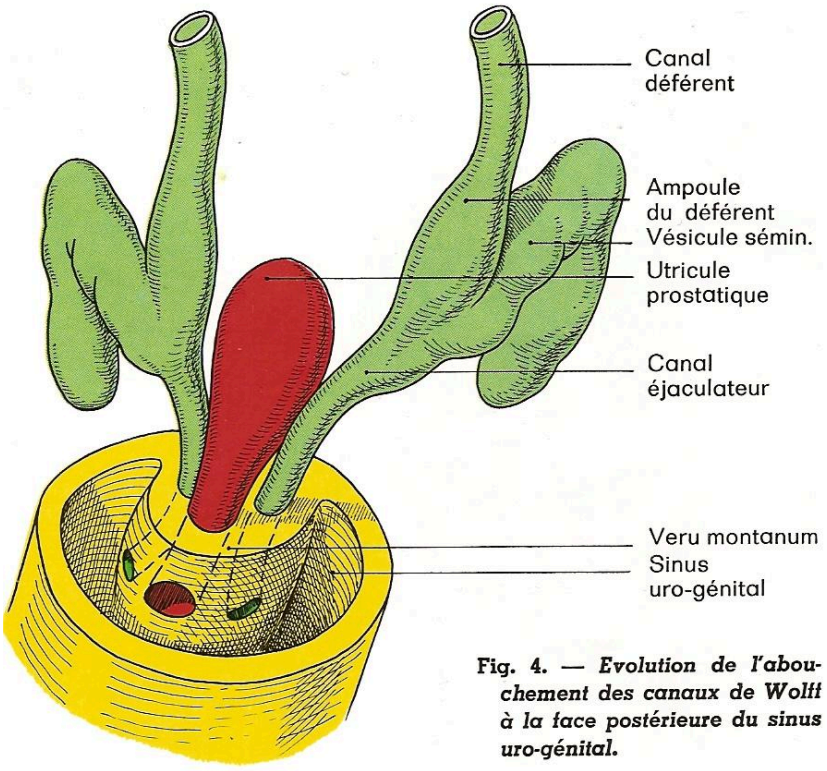


Fig. 4. — *Evolution de l'abouchement des canaux de Wolff à la face postérieure du sinus uro-génital.*

FIG. — Wolffian ducts

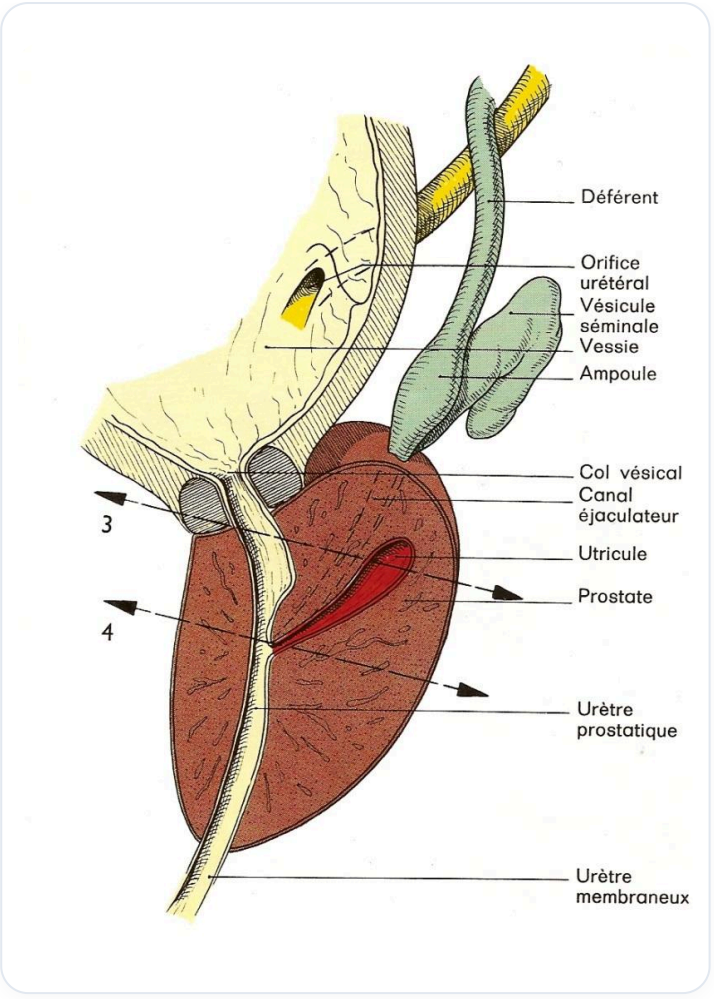


FIG. — Male prostate anatomy

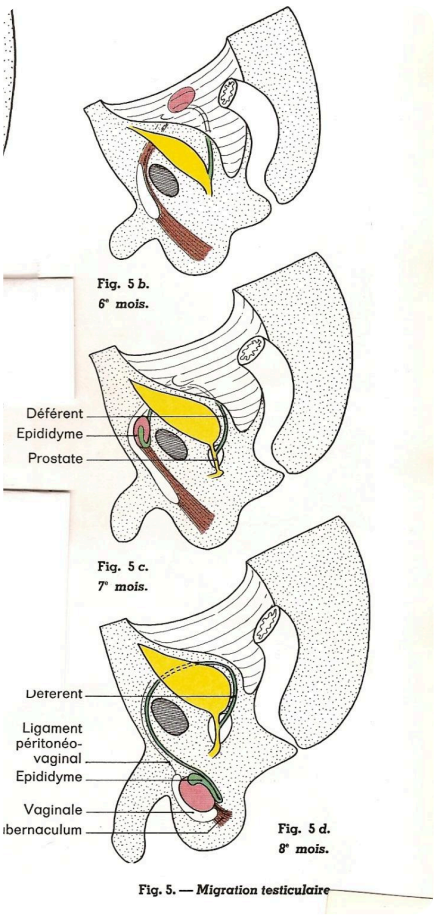


FIG. — Testicular migration